Stat 135 Section

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Life ≫ School ≫ Class Quiz

Figure: *Chutes Montmorency*. Source: All About Canadian History (`http://cdnhistorybits.wordpress.com`)

Models and Inference

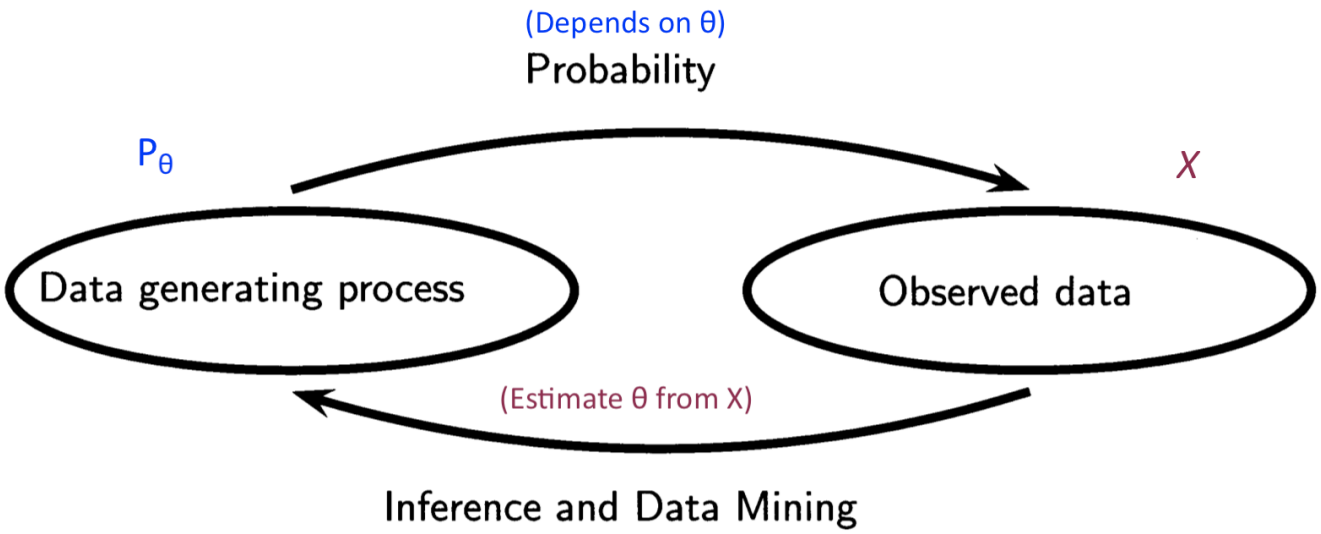

Figure: *Visual statistical inference.* Adapted from: "All of statistics" by Larry Wasserman

Solving MLE Problems

1. Write out the density of the data and treat it as function of parameter θ.

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} P_\theta \Longrightarrow L(\theta; X) = \prod_{i=1}^{n} f_\theta(X_i).$$

2. Optimize the function over θ.

$$\hat{\theta}_{\text{ML}} = \arg\max_{\theta} \{L(\theta; X)\}.$$

   Often, $\ell(\theta; X) \overset{\text{def}}{=} \log L(\theta; X)$ is easier to optimize. Optimize by (i) setting derivative to zero, or (ii) other approaches.

3. The optimizing value, $\hat{\theta}_{\text{ML}}$, of Step 2, is the MLE.

1. Poisson

Consider n i.i.d. random variables, each of them Pois(λ), with unknown parameter λ. Suppose we measure the values $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$. The likelihood function is $\ell(\lambda) = f(k_1|\lambda) \cdots f(k_n|\lambda)$, where $f(k|\lambda) = e^{-\lambda}\lambda^k/k!$, with $k = 0, 1, 2, 3, \ldots$. Find the value of λ that maximizes the probability of the data, i.e., the maximum likelihood estimate $\hat{\lambda}_{\text{ML}}$.

2. Uniform

Consider n i.i.d. continuous random variables, each of them $\text{Unif}(0, \theta)$, with unknown parameter θ. Suppose we measure the values $X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n$ (here $x_i > 0$ for all i). Compute the likelihood function for such data sample, and compute $\hat{\theta}_{\text{ML}}$.

R programming exercise

1. Go to Bcourses page, Pages, lab, and download `beeswax.txt`
2. Load beeswax into R as a dataframe called `bees`
3. Examine the structure of the `bees` using command `str(bees)`
4. Convert columns from type `factor` to type `numeric` whenever
5. Give column names
6. Switch order of columns

R programming exercise

1. Keep just `meltingpoints`>63 and `hydrocarbons`>14 (use `filter()`)
2. Use `mutate()` to multiply every value of hydrocarbon by 2